General transfinite recursion: every rule tau defines a class function sigma with sigma(alpha) = tau restricted to earlier values

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-set-theory-spine/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the definition line, the align computation and the closing coherence remark all appear verbatim with the approximation lemma cited twice by label.

Editorial note. The label transrecursionschema names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own thm environment in the unit source.

Allowed standard prerequisites (editorial).

- Bounded approximation lemma. Same unit, \ollabel{transrecursionfun} (source line 709). Line 749: existence and uniqueness of the alpha-approximation f_alpha for every alpha, and again at line 761 for the coherence identity on overlaps.

Source reference. Original reference transrecursionfun: sources/olp-set-theory-spine/pinned-original-source.tex, source lines 709–709. Prerequisite pointer invoked at proof line(s) 752; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Theorem 0.1 (General Recursion)**.** *For any term $\tau(x)$, we can explicitly define a term $\sigma(x)$, such that $\sigma(\alpha) = \tau(\sigma \restriction \alpha)$ for any ordinal α.*

Proof. For each α, by transrecursionfun there is a unique α-approximation, f_α,

for τ. Define $\sigma(\alpha)$ as $f_{+1\alpha}(\alpha)$. Now:

$$\begin{aligned}\sigma(\alpha) &= f_{+1\alpha}(\alpha)\\ &= \tau(f_{+1\alpha} \upharpoonright \alpha)\\ &= \tau(\{\, \langle \beta, f_{+1\alpha}(\beta)\rangle \mid \beta \in \alpha \,\})\\ &= \tau(\{\, \langle \beta, f_{+1\beta}(\beta)\rangle \mid \beta \in \alpha \,\})\\ &= \tau(\sigma \upharpoonright \alpha)\end{aligned}$$

noting that $f_{+1\beta}(\beta) = f_{+1\alpha}(\beta)$ for all $\beta < \alpha$, as in transrecursionfun. □